Sommersemester 2026

1. Exercise Sheet Solutions

TUTORIUM: Friday, 20.03.2026.

SECTION

01

Getting familiar with the Density of States

The calculation of thermodynamic quantities, response functions and Feynman diagrams in QFT for condensed matter systems often requires the evaluation of integrals or sums over all momenta $\boldsymbol{k}$ (typically over the first Brillouin Zone). An important simplification of these $\boldsymbol{k}$-summations is possible, however, when the integrand $\mathcal{F}$ depends on the **energy only**. In this case the integration/sum is best performed by using the energy ε as a variable. In the case of a cubic lattice of volume L^d in d dimensions, for a given observable F, we have:

$$F = \frac{1}{L^d}\sum_{\boldsymbol{k}} \mathcal{F}(\varepsilon_{\boldsymbol{k}}) = \frac{1}{(2\pi)^d}\frac{(2\pi)^d}{L^d}\sum_{\boldsymbol{k}} \mathcal{F}(\varepsilon_{\boldsymbol{k}}) = \int \mathrm{d}\varepsilon\, \mathcal{N}(\varepsilon)\mathcal{F}(\varepsilon),$$

or, for the continuous case,

$$F = \frac{1}{(2\pi)^d}\int \mathrm{d}^d k\, \mathcal{F}(\varepsilon_{\boldsymbol{k}}) = \int \mathrm{d}\varepsilon\, \mathcal{N}(\varepsilon)\mathcal{F}(\varepsilon),$$

where $\mathcal{N}(\varepsilon)$, i.e. the so-called Density of States (DOS), can be defined by comparison of the different expressions as

$$\mathcal{N}(\varepsilon) = \frac{1}{L^d}\sum_{\boldsymbol{k}} \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}),$$

or, for the continuous case,

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d}\int \mathrm{d}^d k\, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}).$$

a)

PROBLEM

Consider the two cases of particles which can move freely and particles whose motion is bound to an infinite lattice with lattice spacing a. Which of the above expressions do you have to use in the first and second case, respectively? Is the integral/summation restricted to certain $\boldsymbol{k}$-vectors? How does the result change, if one considers a one-dimensional, finite lattice (N lattice points, lattice spacing a) with periodic boundary conditions?

SOLUTION

Freely moving particles can have arbitrary momenta, so one integrates over all of momentum space:

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}).$$

Particles bound to an infinite lattice with lattice spacing a can only have momenta within the first Brillouin zone:

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \int_{[-\frac{\pi}{a}, \frac{\pi}{a}]^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}).$$

Finally, for a one-dimensional finite lattice (N lattice points, spacing a) with periodic boundary conditions, the sum runs over a discrete set of allowed momenta:

$$\mathcal{N}(\varepsilon) = \frac{1}{N} \sum_{k \in K} \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}),$$

with $K = \left\{ -\frac{\pi}{a} + \frac{2\pi}{Na} j \mid j = 0, \ldots, N-1 \right\}$.

b)

PROBLEM

Calculate and plot the explicit expression for $\mathcal{N}(\varepsilon)$ for free, non-interacting particles of mass m (so that $\varepsilon_{\boldsymbol{k}} = \frac{\hbar^2 k^2}{2m}$) in one, two and three dimensions. How do the corresponding Fermi surfaces look like in these cases?

SOLUTION

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}),$$

with $\varepsilon_{\boldsymbol{k}} = \frac{\hbar^2 k^2}{2m}$. For a given energy $\varepsilon > 0$, the corresponding constant-energy shell is a sphere of radius

$$k_\varepsilon = \frac{\sqrt{2m\varepsilon}}{\hbar}.$$

Using

$$\int_{\mathbb{R}^d} \mathrm{d}^d k \, \delta(g(\boldsymbol{k})) = \int_{g^{-1}(0)} \frac{1}{|\boldsymbol{\nabla} g|} \, \mathrm{d}\sigma(k),$$

which for $g(\boldsymbol{k}) = \varepsilon - \varepsilon_{\boldsymbol{k}}$, $g^{-1}(0) = S^{d-1}(k_\varepsilon)$ and $|\boldsymbol{\nabla} g| = \frac{\hbar^2 k}{m}$ yields

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \left| S^{d-1}(k_\varepsilon) \right| \frac{1}{\frac{\hbar^2 k_\varepsilon}{m}},$$

where $S^{d-1}(k)$ denotes the radius-k sphere in d dimensions. At $\varepsilon = \varepsilon_{\mathrm{F}}$, k_ε becomes the Fermi momentum k_{F}, and the constant-energy shell becomes the Fermi surface.

- $d = 1$: $|S^0(k)| = 2$, hence

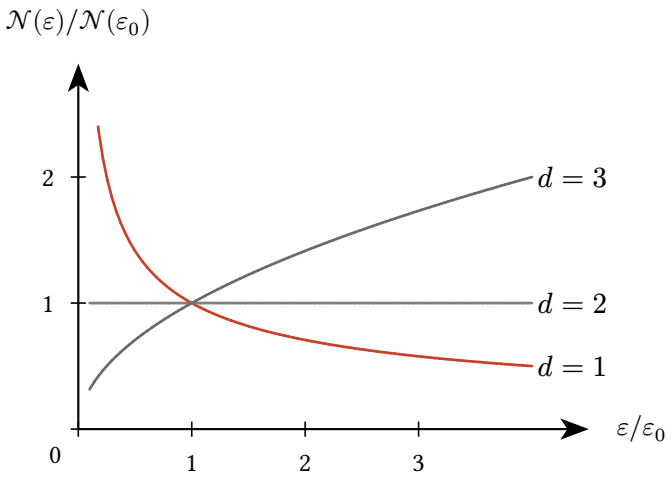
$$\mathcal{N}(\varepsilon) = \frac{1}{2\pi}\frac{\sqrt{2m}}{\hbar\sqrt{\varepsilon}}.$$

- $d = 2$: $|S^1(k)| = 2\pi k$, hence

$$\mathcal{N}(\varepsilon) = \frac{1}{2\pi}\frac{m}{\hbar^2}.$$

- $d = 3$: $|S^2(k)| = 4\pi k^2$, hence

$$\mathcal{N}(\varepsilon) = \frac{1}{2\pi^2}\frac{m}{\hbar^2}\frac{\sqrt{2m\varepsilon}}{\hbar}.$$



At $\varepsilon = \varepsilon_F$, these constant-energy shells become the Fermi surfaces: two points in $d = 1$, a circle in $d = 2$, and a sphere in $d = 3$.

c)

PROBLEM

Consider the following one-dimensional tight-binding Hamiltonian

$$H = -t \sum_{\langle i,j\rangle,\sigma} \left(c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}\right).$$

- Here the hopping ($t > 0$) is restricted to neighboring sites, where $c^\dagger_{i,\sigma}$ and $c_{i,\sigma}$ are the creation/annihilation operators for one electron with spin $\sigma = \uparrow, \downarrow$ at the position $x_i = ia$ with $i = 0, 1, ..., N-1$ and a being the lattice spacing. Assuming periodic boundary conditions ($x_0 = x_N$), compute the corresponding eigenenergies, e.g. using the following basis transformation (from real to momentum space)

$$c^\dagger_{k,\sigma} = \frac{1}{\sqrt{N}} \sum_{x_i} e^{-ikx_i} c^\dagger_{i,\sigma},$$

for the fermionic operators.

SOLUTION

We start from the given basis transformation

$$c^\dagger_{k,\sigma} = \frac{1}{\sqrt{N}} \sum_{x_i} e^{-ikx_i} c^\dagger_{i,\sigma}$$

and invert it to express the real-space operators in terms of momentum-space operators:

$$c^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}} \sum_{k} e^{ikx_i} c^\dagger_{k,\sigma}$$

$$c_{i,\sigma} = \frac{1}{\sqrt{N}} \sum_{k} e^{-ikx_i} c_{k,\sigma}.$$

For periodic boundary conditions, the allowed momenta are

$$k = -\frac{\pi}{a} + \frac{2\pi}{Na} n, \quad n = 0, ..., N-1.$$

With this discrete set of momenta, the finite geometric sum in the Hamiltonian becomes the Kronecker delta appearing below. Substituting this into the Hamiltonian, we obtain (here $j+1$ is understood modulo N because of the periodic boundary conditions)

$$\begin{aligned}
H &= -t\sum_{j,\sigma}\left(c_{j,\sigma}^\dagger c_{j+1,\sigma} + \text{h.c.}\right) \\
&= -t\sum_{j,\sigma}\left(\frac{1}{\sqrt{N}}\sum_k e^{ikx_j}c_{k,\sigma}^\dagger \frac{1}{\sqrt{N}}\sum_{k'} e^{-ik'x_{j+1}}c_{k',\sigma} + \text{h.c.}\right) \\
&= -t\frac{1}{N}\sum_{k,k'}\sum_{j,\sigma}\left(e^{ikja-ik'(j+1)a}c_{k,\sigma}^\dagger c_{k',\sigma} + \text{h.c.}\right),
\end{aligned}$$

Using the identity

$$\frac{1}{N}\sum_j e^{ikja-ik'(j+1)a} = e^{-ik'a}\frac{1}{N}\sum_j e^{ija(k-k')} = e^{-ik'a}\delta_{k,k'},$$

we find

$$\begin{aligned}
H &= -t\sum_{k,k'}\sum_\sigma\left(e^{-ik'a}\delta_{k,k'}c_{k,\sigma}^\dagger c_{k',\sigma} + \text{h.c.}\right) \\
&= -t\sum_{k,\sigma}\left(e^{-ika}c_{k,\sigma}^\dagger c_{k,\sigma} + \text{h.c.}\right) \\
&= -t\sum_{k,\sigma}\left(e^{-ika}c_{k,\sigma}^\dagger c_{k,\sigma} + e^{ika}c_{k,\sigma}^\dagger c_{k,\sigma}\right) \\
&= \sum_{k,\sigma}\varepsilon_k c_{k,\sigma}^\dagger c_{k,\sigma}
\end{aligned}$$

Thus the eigenenergies are

$$\varepsilon_k = -2t\cos(ka)$$

and the corresponding eigenstates are

$$|(k,\sigma)\rangle = c_{k,\sigma}^\dagger|0\rangle.$$

d)

PROBLEM

How can one extend the results of 1c) to arbitrary dimensions $d > 1$? Plot numerically the DOS $\mathcal{N}(\varepsilon)$ for the cases $d = 1, 2, 3$ with $\hbar = m = t = a = 1$. Which are the most prominent features of these DOS functions and at which energies ε do they occur? How would the corresponding Fermi surfaces look like for the cases $d = 1, 2$, e.g. if one has an average density of one electron per site (half-filled system)?

SOLUTION

In arbitrary dimensions $d > 1$, the tight-binding Hamiltonian is

$$H = \sum_{\boldsymbol{k},\sigma}\varepsilon_{\boldsymbol{k}}c_{\boldsymbol{k},\sigma}^\dagger c_{\boldsymbol{k},\sigma},$$

with dispersion

$$\varepsilon_{\boldsymbol{k}} = -2t \sum_{i=1}^{d} \cos(k_i a),$$

The density of states is therefore

$$\mathcal{N}_d(\varepsilon) = \frac{1}{(2\pi)^d} \int_{\left[-\frac{\pi}{a}, \frac{\pi}{a}\right]^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}),$$

Using

$$\int_{\left[-\frac{\pi}{a}, \frac{\pi}{a}\right]^d} \mathrm{d}^d k \, \delta(g(\boldsymbol{k})) = \int_{g^{-1}(0) \cap \left[-\frac{\pi}{a}, \frac{\pi}{a}\right]^d} \frac{\mathrm{d}\sigma(\boldsymbol{k})}{|\boldsymbol{\nabla} g|},$$

with $g(\boldsymbol{k}) = \varepsilon - \varepsilon_{\boldsymbol{k}}$, and setting $\hbar = m = t = a = 1$, we obtain

$$\boldsymbol{\nabla} g = -2 \begin{pmatrix} \sin(k_1) \\ \vdots \\ \sin(k_d) \end{pmatrix}, \quad |\boldsymbol{\nabla} g| = 2 \sqrt{\sum_{i=1}^{d} \sin^2(k_i)},$$

- $d = 1$: For $|\varepsilon| < 2$, there are two solutions $\pm k_\varepsilon$ with

  $$\cos(k_\varepsilon) = -\frac{\varepsilon}{2},$$

  Hence

  $$|\boldsymbol{\nabla} g| = 2|\sin(k_\varepsilon)| = 2\sqrt{1 - \cos^2(k_\varepsilon)} = \sqrt{4 - \varepsilon^2},$$

  and therefore

  $$\mathcal{N}_1(\varepsilon) = \begin{cases} \dfrac{1}{\pi\sqrt{4 - \varepsilon^2}} & \text{if } |\varepsilon| < 2 \\ 0 & \text{otherwise} \end{cases}.$$

- $d = 2$: We write

  $$\mathcal{N}_2(\varepsilon) = \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \mathrm{d}k_x \int_{-\pi}^{\pi} \mathrm{d}k_y \, \delta\big(\varepsilon + 2\cos(k_x) + 2\cos(k_y)\big),$$

  For fixed k_x, define

  $$A = \varepsilon + 2\cos(k_x),$$

  Then the inner integral is exactly the $d = 1$ result:

  $$\int_{-\pi}^{\pi} \mathrm{d}k_y \, \delta\big(A + 2\cos(k_y)\big) = \frac{2}{\sqrt{4 - A^2}}.$$

  Thus

  $$\mathcal{N}_2(\varepsilon) = \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \mathrm{d}k_x \, \frac{2}{\sqrt{4 - (\varepsilon + 2\cos(k_x))^2}},$$

  With $u = \cos(k_x)$ and symmetry, this becomes

$$\mathcal{N}_2(\varepsilon) = \frac{1}{\pi^2} \int_{-1}^{1} \frac{\mathrm{d}u}{\sqrt{1-u^2}\sqrt{4-(\varepsilon+2u)^2}}.$$

This is a complete elliptic integral of the first kind, so

$$\mathcal{N}_2(\varepsilon) = \begin{cases} \frac{1}{2\pi^2} K\left(\sqrt{1-\frac{\varepsilon^2}{16}}\right) & \text{if } |\varepsilon| < 4 \\ 0 & \text{otherwise} \end{cases},$$

- $d = 3$: Starting from

$$\mathcal{N}_3(\varepsilon) = \frac{1}{(2\pi)^3} \int_{-\pi}^{\pi} \mathrm{d}k_x \int_{-\pi}^{\pi} \mathrm{d}k_y \int_{-\pi}^{\pi} \mathrm{d}k_z \, \delta\big(\varepsilon + 2\cos(k_x) + 2\cos\big(k_y\big) + 2\cos(k_z)\big),$$

we fix k_z and define

$$\varepsilon' = \varepsilon + 2\cos(k_z),$$

The remaining k_x, k_y integral is exactly the $d = 2$ DOS at energy ε':

$$\mathcal{N}_3(\varepsilon) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \mathrm{d}k_z \, \mathcal{N}_2(\varepsilon + 2\cos(k_z)),$$

Inserting the $d = 2$ result gives

$$\mathcal{N}_3(\varepsilon) = \frac{1}{4\pi^3} \int_{-\pi}^{\pi} \mathrm{d}k_z \, K\left(\sqrt{1-\frac{(\varepsilon+2\cos(k_z))^2}{16}}\right),$$

where the integrand contributes only if

$$|\varepsilon + 2\cos(k_z)| < 4.$$

Equivalently, with $u = \cos(k_z)$,

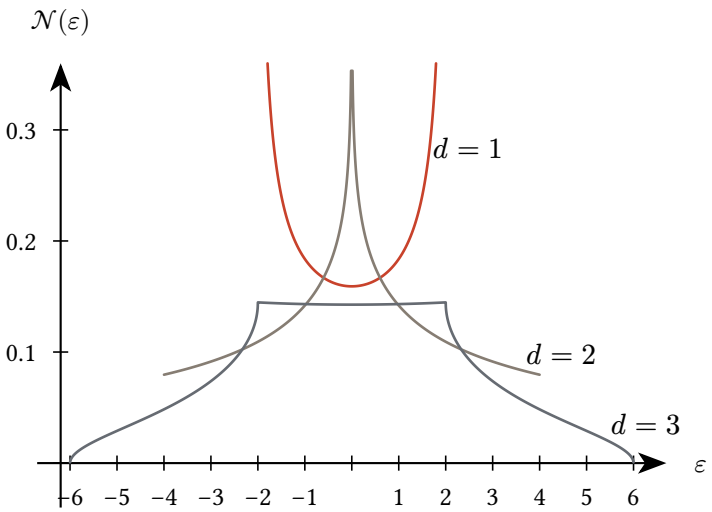
$$\mathcal{N}_3(\varepsilon) = \frac{1}{2\pi^3} \int_{-1}^{1} \frac{K\left(\sqrt{1-\frac{(\varepsilon+2u)^2}{16}}\right)}{\sqrt{1-u^2}} \, \mathrm{d}u,$$

again with the restriction

$$|\varepsilon + 2u| < 4.$$

In particular,

$$\mathcal{N}_3(\varepsilon) = 0 \quad \text{for} \quad |\varepsilon| > 6.$$



Prominent Features of the DOS Functions

- $d = 1$: inverse square root divergence at the band edges $\varepsilon = \pm 2$.
- $d = 2$: logarithmic van Hove singularity at $\varepsilon = 0$, finite band edges at $\varepsilon = \pm 4$.
- $d = 3$: non-analytic kinks at $\varepsilon = \pm 2$, band edges at $\varepsilon = \pm 6$. The kinks occur because at $\varepsilon = \pm 2$ the topology of the constant-energy surface changes, so the phase-space measure is non-analytic there even though the DOS remains finite.

For one electron per site (half filling), exactly half of all available single-particle states are occupied. Since

$$\varepsilon_{\boldsymbol{k}} = -2 \sum_{i=1}^{d} \cos(k_i),$$

is particle-hole symmetric on the bipartite hypercubic lattice, the spectrum is symmetric around $\varepsilon = 0$. Therefore at half filling the Fermi energy at $T = 0$ is

$$\varepsilon_{\mathrm{F}} = 0,$$

The Fermi surface is defined by $\varepsilon_{\boldsymbol{k}} = \varepsilon_{\mathrm{F}}$, i.e.

$$-2 \sum_{i=1}^{d} \cos(k_i) = 0,$$

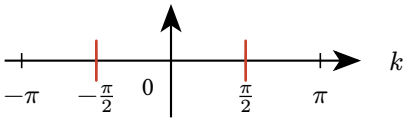
Therefore

$$d = 1: \quad \cos(k) = 0 \quad \rightarrow \quad k = \pm \frac{\pi}{2},$$

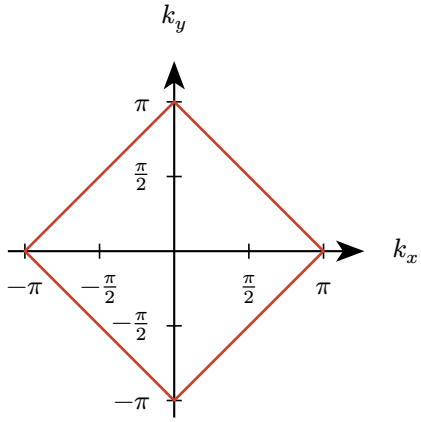
and

$$d = 2: \quad \cos(k_x) + \cos(k_y) = 0,$$

i.e. a diamond-shaped Fermi surface in the first Brillouin zone.



Fermi surface for $d = 1$ at half filling



Fermi surface for $d = 2$ at half filling

SECTION

02

Screened and Unscreened Coulomb Potentials

a)

PROBLEM

From the integral representation of the delta function,

$$\delta(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}},$$

and the fact that the Coulomb potential $\phi(\boldsymbol{r}) = -e/r$ satisfies Poisson's equation,

$$-\nabla^2 \phi(\boldsymbol{r}) = -4\pi e \delta(\boldsymbol{r}),$$

show that the electronic pair potential, $V(\boldsymbol{r}) = -e\phi(\boldsymbol{r}) = e^2/r$, can be written in the form

$$V(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} V(\boldsymbol{k}),$$

where the Fourier transform $V(\boldsymbol{k})$ is given by

$$V(\boldsymbol{k}) = \frac{4\pi e^2}{k^2}.$$

SOLUTION

Assume that the Coulomb potential has the Fourier representation

$$\phi(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \phi(\boldsymbol{k}),$$

Applying $-\nabla^2$ gives

$$-\nabla^2\phi(\boldsymbol{r}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}(-\nabla^2 e^{i\boldsymbol{k}\cdot\boldsymbol{r}})\phi(\boldsymbol{k}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} k^2\phi(\boldsymbol{k}),$$

since

$$-\nabla^2 e^{i\boldsymbol{k}\cdot\boldsymbol{r}} = k^2 e^{i\boldsymbol{k}\cdot\boldsymbol{r}}.$$

On the other hand, Poisson's equation implies

$$-\nabla^2\phi(\boldsymbol{r}) = -4\pi e\delta(\boldsymbol{r}) = -4\pi e \int \frac{\mathrm{d}^3k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}},$$

Comparing the Fourier coefficients of $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$ gives

$$k^2\phi(\boldsymbol{k}) = -4\pi e,$$

hence

$$\phi(\boldsymbol{k}) = -\frac{4\pi e}{k^2}.$$

Therefore

$$V(\boldsymbol{k}) = -e\phi(\boldsymbol{k}) = \frac{4\pi e^2}{k^2}.$$

Thus

$$V(\boldsymbol{r}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} V(\boldsymbol{k}),$$

with

$$V(\boldsymbol{k}) = \frac{4\pi e^2}{k^2}.$$

b)

PROBLEM

Show that the Fourier transform of the screened Coulomb interaction $V_s(\boldsymbol{r}) = (e^2/r)e^{-k_{\mathrm{TF}}r}$ is

$$V_s(\boldsymbol{k}) = \frac{4\pi e^2}{k^2 + k_{\mathrm{TF}}^2},$$

by substituting this expression into the Fourier integral

$$V_s(\boldsymbol{r}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} V_s(\boldsymbol{k}),$$

and evaluating that integral in spherical coordinates (Hint: The radial integral is best done as a contour integral.). Finally, deduce that $V_s(\boldsymbol{r})$ satisfies

$$(-\nabla^2 + k_{\mathrm{TF}}^2)V_s(\boldsymbol{r}) = 4\pi e^2\delta(\boldsymbol{r}).$$

Following the hint, we substitute the Fourier transform of the screened Coulomb interaction into the Fourier integral:

$$\begin{aligned} V_s(\boldsymbol{r}) &= \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} V_s(\boldsymbol{k}) \\ &= \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \frac{4\pi e^2}{k^2 + k_{\mathrm{TF}}^2}, \end{aligned}$$

and pass to spherical coordinates:

$$\begin{aligned} V_s(\boldsymbol{r}) &= \int \frac{k^2 \sin\theta \,\mathrm{d}k \,\mathrm{d}\theta \,\mathrm{d}\varphi}{(2\pi)^3} e^{ikr\cos\theta} \frac{4\pi e^2}{k^2 + k_{\mathrm{TF}}^2} \\ &= \int \frac{k^2 \sin\theta \,\mathrm{d}k \,\mathrm{d}\theta}{(2\pi)^2} e^{ikr\cos\theta} \frac{4\pi e^2}{k^2 + k_{\mathrm{TF}}^2}. \end{aligned}$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}\theta} e^{ikr\cos\theta} = -e^{ikr\cos\theta} ikr \sin\theta$$

we can write

$$V_s(\boldsymbol{r}) = \int \frac{k \,\mathrm{d}k \,\mathrm{d}\theta}{(2\pi)^2} \left(\frac{\mathrm{d}}{\mathrm{d}\theta} e^{ikr\cos\theta} \right) \frac{1}{-ir} \frac{4\pi e^2}{k^2 + k_{\mathrm{TF}}^2}$$

and then carry out the θ-integral:

$$\int_0^\pi \mathrm{d}\theta \left(\frac{\mathrm{d}}{\mathrm{d}\theta} e^{ikr\cos\theta} \right) = e^{ikr\cos\pi} - e^{ikr\cos 0} = e^{-ikr} - e^{ikr} = -2i\sin(kr),$$

yielding

$$V_s(\boldsymbol{r}) = \frac{2e^2}{\pi r} \int_0^\infty \mathrm{d}k \, \frac{k\sin(kr)}{k^2 + k_{\mathrm{TF}}^2}.$$

This integrand is an even function in k, so we can extend it to the whole real axis and write

$$V_s(\boldsymbol{r}) = \frac{e^2}{\pi r} \int_{-\infty}^\infty \mathrm{d}k \, \frac{k\sin(kr)}{k^2 + k_{\mathrm{TF}}^2}.$$

To evaluate the radial integral by contour integration, we instead consider

$$I(r) = \frac{e^2}{\pi r} \int_{-\infty}^\infty \mathrm{d}k \, \frac{k e^{ikr}}{k^2 + k_{\mathrm{TF}}^2},$$

so that $V_s(\boldsymbol{r}) = \mathrm{Im}\ I(r)$.

The integrand of $I(r)$ has poles at $k = \pm i k_{\mathrm{TF}}$. For $r > 0$, the factor e^{ikr} decays exponentially in the upper half-plane, so we close the contour there and compute the residue

$$\mathrm{Res}_{k=ik_{\mathrm{TF}}} \frac{k e^{ikr}}{k^2 + k_{\mathrm{TF}}^2} = \lim_{k\to ik_{\mathrm{TF}}} (k - ik_{\mathrm{TF}}) \frac{k e^{ikr}}{(k - ik_{\mathrm{TF}})(k + ik_{\mathrm{TF}})} = \frac{ik_{\mathrm{TF}} e^{-k_{\mathrm{TF}} r}}{2ik_{\mathrm{TF}}} = \frac{1}{2} e^{-k_{\mathrm{TF}} r}.$$

This gives

$$I(r) = \frac{e^2}{\pi r} 2\pi i \mathrm{Res}_{k=ik_{\mathrm{TF}}} \frac{ke^{ikr}}{k^2 + k_{\mathrm{TF}}^2} = \frac{e^2}{r} i e^{-k_{\mathrm{TF}} r}$$

and therefore

$$V_s(\boldsymbol{r}) = \mathrm{Im}\ I(r) = \frac{e^2}{r} e^{-k_{\mathrm{TF}} r}$$

For the second part, we proceed exactly as in part a) and write the differential equation

$$(-\nabla^2 + k_{\mathrm{TF}}^2) V_s(\boldsymbol{r}) = 4\pi e^2 \delta(\boldsymbol{r})$$

in Fourier space as

$$(k^2 + k_{\mathrm{TF}}^2) V_s(\boldsymbol{k}) = 4\pi e^2$$

which is indeed satisfied by

$$V_s(\boldsymbol{k}) = \frac{4\pi e^2}{k^2 + k_{\mathrm{TF}}^2}.$$